



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

**Department of Computer Science
Faculty of Science & Technology (FST)**

Fall 2024-25

Section: H, Group: 01

PROJECT TITLE

**Virtual Classroom Platform: An Interactive, Accessible, and Engaging
Online Learning Environment**

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Submitted By

Group Members

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Project Estimation (COMOMO)

For our Organic system project , let's take the number of lines of code to be,

SLOC =7000,

Effort Factor Coefficient = 2.4,

P= 1.05 and

T = 0.38.

We know ,

Effort = PM = Coefficient_{<Effort Factor>}*(SLOC/1000)^P

Effort (PM) = 2.4 x (7000/1000)^1.05

Effort (PM) =18.5167

Development time = DM = 2.50*(PM)^T

= 2.50 x (18.5167)^0.38

= 7.5790 Months

Required number of people = ST = PM/DM

=18.5167 / 7.5790

=2.449

Human Resource Allocation

Resource Name	Role	Responsibilities	Skill Set Required	Allocation (Hours/Week)	Duration
Project Manager	Manager	Plan, monitor, and control project activities, ensure risk management, and QA	Leadership, Communication, PM Tools	20	01/01/2025 - 03/31/2025
Business Analyst	Analyst	Gather requirements, interact with stakeholders, and document user needs	Communication, Analytical, Documentation	15	01/01/2025 - 01/31/2025
UI/UX Designer	Designer	Design platform layout, create interactive prototypes, and improve user experience	Figma, Adobe XD, Creativity	25	02/01/2025 - 02/28/2025
Frontend Developer	Developer	Build UI components, connect APIs, and ensure responsiveness	React.js, HTML/CSS, JavaScript	30	03/01/2025 - 05/31/2025
Backend Developer	Developer	Develop APIs, authentication, and logic for backend services	Node.js, Python, Database Management	30	03/01/2025 - 05/31/2025
Database Administrator	Administrator	Design, implement, and optimize databases	SQL, MongoDB, Backup/Recovery Skills	20	03/01/2025 - 05/15/2025
Quality Analyst	Tester	Create and execute test cases, conduct UAT, and debug	Test Automation, Problem-Solving	25	05/01/2025 - 06/15/2025

Deployment Engineer	Engineer	Setup hosting, deploy application, and manage configurations	AWS, Azure, CI/CD	20	06/01/2025 - 06/15/2025
Trainer	Educator	Conduct training for users and provide documentation	Communication, Training, Documentation	10	06/15/2025 - 06/30/2025
Technical Support Staff	Support	Monitor platform, resolve user issues, and provide updates	Troubleshooting, Communication	10	07/01/2025 - Ongoing

Environment Resource Allocation

Environment Type	Purpose	Resources Required	Allocated Duration	Key Dependencies
Development Environment	For coding, debugging, and integrating components	Workstations, IDEs (e.g., Visual Studio, IntelliJ), GitHub repository, test servers	January 2025 - May 2025	Version Control, Developer Access
Testing Environment	For conducting unit, integration, and user acceptance testing	Test servers, automated testing tools (e.g., Selenium, JMeter), staging databases	April 2025 - June 2025	QA Team Access, Test Data
Staging Environment	Simulates production for final validation before deployment	Staging servers, replica database, network configuration	May 2025 - June 2025	Final Build, Deployment Approval
Production Environment	Live platform for end-users	AWS EC2 instances, S3 storage, SSL certificate, Load balancers	June 2025 - Ongoing	Scheduled Backup Configuration
Backup Environment	For data backups and	Cloud storage, Backup software (e.g., AWS	March 2025 - Ongoing	Trainer Access, Training Materials

	disaster recovery	Backup, Azure Backup)		
Training Environment	For user training and simulation	simulation Training servers, replica platform instance, dummy user accounts	June 2025	Trainer Access, Training Materials

Re-usable Software Resource Allocation

Resource Name	Purpose	Modules/Tasks Used	Licensing/Ownership	Allocated Duration
Node.js	Backend development and API creation	Backend Development (Authentication, APIs)	Open Source	January 2025 - May 2025
React.js	Frontend development for user interface	Dashboard, Chat Interface, Meeting UI	Open Source	January 2025 - May 2025
MySQL	Database management	User Data, Assignments, Exams	Open Source	January 2025 - Ongoing
Figma	UI/UX design prototyping	prototyping Login, Dashboard, Calendar Design	Freemium	January 2025 - March 2025
AWS Cloud Services	Hosting and storage	Production Environment	Pay-as-you-go	May 2025 - Ongoing
Jest	Unit testing for JavaScript code	Frontend and Backend Unit Tests	Open Source	April 2025 - May 2025
Bootstrap/Material UI	UI design framework	Frontend Components (Forms, Buttons)	Open Source	February 2025 - May 2025

Risk Analysis:

Impact		Very Low	Low	Medium	High	Very high				
Probability	Very high	Moderate	Severe	Severe	Critical	Critical				
	High	Sustainable	Moderate	Severe	Critical	Critical				
	Medium	Sustainable	Moderate	Moderate	Severe	Critical			Probability	Impact
	Low	Sustainable	Sustainable	Moderate	Severe	Critical			Very high	Very high
	Very Low	Sustainable	Sustainable	Sustainable	Moderate	Severe			High	High
									Medium	Medium
									Low	Low
									Very Low	Very Low

Unique id	Risk description	Caused by	Risk owner	Probability	Impact	Risk rating	Controls	Controls owners	Residual probability	Residual Impact	Residual rating
	Bugs in critical functionalities	Software defects	Development Team	High	Medium	Moderate	Robust testing process	QA Team	Medium	Medium	Moderate
	Performance issues during peak usage	System overload	Development Team	High	High	Critical	Load testing, scalable architecture	DevOps Team	Medium	Medium	Severe
	Scope creep	Changing user requirements	Project Manager	Medium	High	Severe	Clear scope, change management protocols	Project Manager	Low	Medium	Moderate
	Unauthorized access to sensitive data	Security vulnerabilities	Security Team	Medium	Very High	Critical	Data encryption, MFA, penetration testing	Security Team	Low	High	Severe
	Low adoption rate	User unfamiliarity	Product Manager	Medium	Medium	Moderate	User training, intuitive design	Training Team	Low	Low	Sustainable
	Budget overruns	Underestimated costs	Product Manager	Medium	Medium	Moderate	Contingency funds, expense tracking	Finance Team	Low	Medium	Moderate
	Delays in resource allocation	Resource unavailability	HR Team	Medium	High	Severe	Flexible timelines, resource backups	Project Manager	Low	Medium	Moderate
	Data loss due to system crashes	Hardware/software failures	IT Team	Low	High	Moderate	Regular backups, recovery systems	IT Team	Low	Low	Sustainable
	Negative feedback on usability	Poor user experience	Design Team	Medium	Medium	Moderate	Beta testing, incorporate user feedback	UX/UI Team	Low	Medium	Moderate
	Post-launch system downtime	Maintenance errors	DevOps Team	Low	Medium	Sustainable	Monitoring tools, dedicated support team	DevOps Team	Very Low	Medium	Sustainable